A FAI REPORT ON

UI/UX DESIGN AND PROTOTYPING OF

“FINWISE” EXPENSE TRACKER MOBILE APPLICATION

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[Sem-II, A.Y. 2025-26]

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'येथे बहुतांचे हित ।'

CERTIFICATE

This is to certify that Sarthak S. Hiwarde : B25IT1094 , Atharva S. Jagtap : B25IT1094, Atharva V. Kakade : B25IT1101, Parthraj V. Malave : B25IT1106, have successfully completed the PBL Project entitled "Mobile Applicatio Design:**FINWISE**." under my supervision, in the partial fulfillment of First Year Engineering of Savitribai Phule Pune University.

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ACKNOWLEDGEMENT

First of all, we express our deep gratitude to our project guide Prof. V.V. Parode for his valuable support, help & guidance from time to time during the project work. We are also grateful to our Head of Department, Dr. A.S. Sawaikar for giving us this opportunity to present this project report. Last but not the least; we would like to thank our entire teaching and Non-teaching staff who assisted us directly or indirectly throughout the duration of this project.

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ABSTRACT

In the rapidly evolving digital era, effective financial management has become essential for individuals, particularly students and young professionals. Despite the availability of numerous expense tracking applications, many users face difficulties due to complex interfaces, cluttered dashboards, and lack of intuitive navigation. This project focuses on the design and prototyping of a user-friendly mobile expense tracker application using Figma, applying core UI/UX design principles and user-centered methodologies.

The project began with comprehensive user research including surveys and stakeholder interviews to identify user pain points and expectations. Based on research findings, a simplified interface structure was conceptualized, emphasizing clarity, minimalism, and accessibility. The application includes features such as categorized expense tracking, dashboard overview, graphical analytics (daily, weekly, monthly), calendar integration, search functionality, and profile management.

The design employs a green-themed visual identity representing financial growth and stability. Structured navigation, card-based layouts, visual hierarchy, and consistent spacing were implemented to enhance usability and reduce cognitive load. Interactive high-fidelity prototypes were developed and evaluated through usability considerations.

This project demonstrates the practical application of user-centered design, interaction design principles, and visual communication techniques in developing an intuitive financial tracking interface. The prototype serves as a foundation for future backend implementation and advanced financial analytics integration.

Keywords: UI/UX Design, Expense Tracker, Figma Prototype, Financial Dashboard, User-Centered Design, Mobile Interface Design.

1. INTRODUCTION

In the modern digital world, financial management has become a critical aspect of everyday life. With the increasing use of smartphones and digital transactions, individuals are spending money more frequently but often without maintaining structured records. This leads to poor budgeting, overspending, and lack of financial awareness.

Expense tracking applications have emerged as a solution to this problem. However, many existing applications are either overly complex or visually cluttered, making them difficult for students and young professionals to use consistently. Therefore, there is a need for a simple, intuitive, and visually structured financial tracking interface.

This project focuses on designing and prototyping a mobile-based expense tracker application using Figma. The application aims to provide users with an organized dashboard, categorized expense management, and graphical analysis tools to help them understand their spending patterns efficiently.

The project follows a user-centered design approach involving research, requirement analysis, ideation, prototyping, and usability evaluation.

1.1 Background of the Study

Financial literacy among students and early-career professionals is often limited. Many individuals do not track small daily expenses such as food, transport, and shopping. Over time, these small expenses accumulate into significant monthly spending without users realizing it.

Traditional methods such as manual record-keeping or spreadsheet tracking are time-consuming and not user-friendly. While mobile apps provide a better alternative, many of them include excessive features, complex navigation systems, and overwhelming dashboards that discourage regular use.

Research indicates that users prefer:

- Quick expense entry
- Clear visual summaries
- Category-based classification
- Minimal navigation complexity

Therefore, designing a simplified, visually clean, and accessible expense tracking interface becomes essential.

The background study emphasizes the importance of simplicity, usability, and structured information presentation in financial applications.

1.2 About the Project

The project titled **"UI/UX Design and Prototyping of an Expense Tracker Application"** is developed as part of the Project-Based Learning (PBL) curriculum.

The application is designed using Figma and includes high-fidelity interactive screens. It focuses primarily on frontend interface design and user experience rather than backend implementation.

The major modules included in the application are:

- Splash Screen
- Login & Signup Screens
- Dashboard with Total Balance Overview
- Category-based Expense Cards (Food, Rent, Medicine, Transport, Grocery)
- Add Expense Interface
- Transaction History Screen
- Daily / Weekly / Monthly Analysis Toggle
- Pie Chart and Bar Graph Visualization
- Calendar View
- Search Functionality
- Profile and Security Settings

The interface uses a green color theme symbolizing growth, balance, and financial stability. A bottom navigation structure ensures ease of access and thumb-friendly usability.

The prototype demonstrates structured navigation flow and interactive transitions between screens.

1.3 Objectives

The primary objective of this project is to design a user-friendly and visually structured expense tracking application.

The specific objectives include:

1. To understand user needs related to expense management.
2. To design a clean and intuitive mobile interface.
3. To implement category-based expense classification.
4. To provide graphical representations of spending patterns.
5. To ensure consistency in layout and navigation.
6. To apply UI/UX design principles such as hierarchy, contrast, alignment, and proximity.
7. To develop an interactive high-fidelity prototype using Figma.
8. To evaluate usability and improve the design based on feedback.

1.4 Scope of the Project

The scope of this project includes:

- Conducting user research and surveys
- Identifying functional and non-functional requirements
- Designing wireframes and high-fidelity UI screens
- Creating an interactive prototype
- Performing usability evaluation

The project does not include:

- Backend database integration
- Real-time cloud synchronization
- Payment gateway implementation
- Deployment on Android or iOS app stores

Future scope of the project may include:

- AI-based expense prediction
- Budget goal tracking
- Multi-device synchronization
- Data export functionality

2. UNDERSTANDING USER NEEDS

Understanding user needs is a critical step in user-centered design. Before developing the interface, it was necessary to identify problems faced by potential users and understand their expectations.

User research was conducted to gather insights about:

- Current expense tracking habits
- Difficulties in existing applications
- Preferred features
- Visualization preferences

The findings from research directly influenced design decisions and feature prioritization.

2.1 Survey Design & Methodology

A structured survey questionnaire was prepared to collect quantitative and qualitative data from potential users.

The survey targeted:

- Engineering students
- Young professionals
- Smartphone users aged between 18–30 years

The questionnaire included:

- Multiple-choice questions

- Likert scale rating questions
- Open-ended descriptive questions

The survey focused on:

- Frequency of expense tracking
- Preferred analysis method (graphs or numbers)
- Challenges faced in current apps
- Desired features in a new application

The survey was conducted online to ensure easy participation and accessibility.

2.2 Data Collection

Data was collected from approximately 40 respondents. The responses were categorized and tabulated for analysis.

Major observations:

- 75% of respondents do not track daily expenses regularly.
- 68% prefer graphical representation over numerical summaries.
- 60% find existing expense tracking apps complicated.
- 82% want faster and simpler expense entry methods.

The collected data highlighted the importance of simplicity, visual clarity, and structured navigation.

2.4 Functional Requirements

Functional requirements define the specific operations and services that the application must provide to users.

The proposed expense tracker application shall perform the following functions:

2.3 Data Analysis

The collected responses were analyzed to identify recurring themes and user expectations.

Key findings include:

- Users prefer dashboards showing total balance prominently.
- Category-based expense grouping improves understanding.
- Visual charts enhance decision-making.
- Too many features reduce usability.

Based on this analysis, the application design was simplified to include essential features with clear navigation and minimal complexity.

User Management

1. The system shall allow new users to register using email and password.
2. The system shall allow existing users to log in securely.
3. The system shall provide logout functionality.
4. The system shall allow users to update profile information.
5. The system shall allow users to modify password under security settings.

Dashboard Management

6. The system shall display total available balance.
7. The system shall display categorized expense summaries.
8. The system shall update balance automatically after adding expense.
9. The system shall display quick summary cards (Food, Rent, Medicine, Transport, Grocery).

Expense Management

10. The system shall allow users to add new expenses.
11. The system shall allow users to select expense category.
12. The system shall allow users to enter amount and description.
13. The system shall store date of expense automatically.

14. The system shall allow editing existing transactions.
15. The system shall allow deletion of transactions.
16. The system shall validate input fields before saving.

Transaction History

17. The system shall display chronological transaction history.
18. The system shall allow filtering transactions by category.
19. The system shall allow searching transactions by keyword.
20. The system shall display transaction details upon selection.

Data Analysis & Visualization

21. The system shall provide daily expense analysis.
22. The system shall provide weekly expense analysis.
23. The system shall provide monthly expense analysis.
24. The system shall display graphical representation (bar graph).
25. The system shall display pie chart visualization.
26. The system shall allow toggling between different time frames.

Calendar Integration

27. The system shall display expenses according to selected calendar date.
28. The system shall highlight dates with recorded expenses.

Navigation

29. The system shall provide bottom navigation menu.
30. The system shall allow smooth transition between screens.
31. The system shall provide quick access to Add Expense function.

2.5 Non-Functional Requirements

Non-functional requirements define quality attributes and system performance standards.

Usability

1. The interface shall be intuitive and easy to learn.
2. The app shall require minimal training for first-time users.
3. The navigation shall be consistent across all screens.
4. The system shall follow proper visual hierarchy principles.

Performance

5. Screen transitions shall be smooth and responsive.
6. Graph loading shall not cause visual lag.
7. The app design shall support fast data entry.

Reliability

8. The system shall ensure data accuracy.
9. The system shall prevent accidental deletion through confirmation prompts.

Security

10. Login credentials shall be protected through secure design.
11. Password fields shall mask user input.
12. Sensitive profile settings shall be protected.

Accessibility

13. The interface shall use readable font sizes.
14. Color contrast shall meet accessibility guidelines.
15. Icons shall be easily recognizable.
16. Buttons shall be adequately sized for touch interaction.

Scalability

17. The design shall allow future backend integration.
18. The structure shall support additional modules like budget goals.
19. The interface shall be adaptable for future feature expansion.

Maintainability

20. The UI components shall follow consistent design patterns.
21. The layout shall allow easy updates without redesigning entire interface.

3. STAKEHOLDER INTERVIEWS

3.1 Interview Preparation

- A structured questionnaire was prepared.
- Participants were informed about project purpose.
- Interviews were conducted individually.
- Responses were documented for analysis.

Participants:

- 2 Engineering Students
- 1 Working Professional
- 1 Faculty Member

3.2 Interview Questions and Responses

Question 1:

How do you currently track your expenses?

Answer (Student 1):
"I usually don't track daily expenses. I try to remember them mentally."

Answer (Professional):
"I use a simple notes app but it becomes messy over time."

Question 2:

What problems do you face while managing expenses?

Answer (Student 2):
"I forget small expenses like snacks and transport."

Answer (Faculty):
"It is difficult to analyze where most money is being spent."

Question 3:

What features would you expect in an expense tracking app?

Answer (Student 1):
"A quick add button to enter expense instantly."

Answer (Professional):
"Monthly summary with charts would be helpful."

Question 4:

Do you prefer numbers or graphical representation?

Answer
"Graphs are easier to understand compared to raw numbers."

Question 5:

What makes you stop using an app?

Answer (Student 2):
"If it is too complicated or has too many steps."

Answer (Professional):
"If it takes too much time to enter data."

Question 6:

Do you think category-based tracking is useful?

Answer (All Participants):
"Yes, categories help understand spending habits clearly."

Question 7:

Would you prefer dark mode or light mode?

Answer:
"Light mode with clean design is more comfortable."

3.3 Key Insights Derived

From the interviews, the following key insights were identified:

1. Users prefer quick expense entry.
2. Graphical visualization increases clarity.
3. Simple navigation improves engagement.
4. Category-based classification is essential.
5. Dashboard must clearly display total balance.
6. Minimal steps reduce user frustration.
7. Clean light theme is preferred.

These insights directly influenced the final interface design of the application.

4. DESIGN CONCEPT & IDEATION

The design concept phase involved converting research insights into structured interface ideas. The goal was to create a clean, modern, and financial-themed design.

4.1 Brainstorming Ideas

Multiple design ideas were explored:

- Light theme vs dark theme

- Card layout vs list layout
- Floating action button vs bottom navigation
- Detailed dashboard vs minimal dashboard

After evaluation, a light green-themed card-based layout with bottom navigation was selected for clarity and simplicity.

4.2 Feature Planning

Features were categorized based on priority.

High Priority Features:

- Dashboard Summary
- Add Expense
- Transaction History
- Analysis Charts

Medium Priority Features:

- Calendar View
- Search Function

Future Scope Features:

- Budget Goals
- AI-based suggestions
- Expense reminders

4.3 Final Concept Selection

The final concept includes:

- Green gradient dashboard

- Category-based expense cards
- Interactive analysis charts
- Bottom navigation menu
- Clean typography
- Profile and security settings

The selected design ensures usability, consistency, and clarity while maintaining a modern appearance.

1. User Personas & Journey Mapping

User Persona 1: Student (Budget Tracker)

- **Name:** Rahul
- **Age:** 20
- **Goal:** Track daily expenses and save money
- **Pain Points:**
  - Overspending
  - No clear idea where money goes

Journey:

1. Opens app → sees dashboard
2. Adds income (pocket money)
3. Adds expenses (food, travel, etc.)
4. Views charts & reports
5. Adjusts spending habits

2. Task Flows & Information Architecture

Main Task Flow (Expense Tracking)

Open App
↓
Login / Sign Up
↓
Dashboard

↓
Add Transaction
↓
Select Category
↓
Save
↓
View Reports / Charts

Information Architecture (IA)

Main Sections:

- **Home (Dashboard)**
- **Transactions**
- **Analytics**
- **Budget**
- **Profile / Settings**

3. High-Fidelity UI Design Explanation

Your UI shows a **modern finance app design** with:

Key Screens:

- Splash Screen
- Login / Signup
- Dashboard
- Add Transaction
- Analytics (Charts)
- Calendar View
- Profile Settings

UI Design Features:

- Clean green theme (finance-friendly color)
- Card-based layout for data

- Graphs for visualization
- Bottom navigation bar
- Minimal and modern look

Dashboard Includes:

- Total balance
- Income vs Expense
- Recent transactions
- Quick action buttons

Analytics Screen:

- Bar charts
- Pie charts
- Time-based insights

Add Transaction:

- Category selection
- Amount input
- Date picker
- Notes

4. Style Guide Development

Colors:

- Primary: Green (#00B894 or similar)
- Secondary: Light Green / Mint
- Background: White / Light Grey
- Accent: Blue (for charts)

Typography:

- Headings: Bold, clean font
- Body: Simple sans-serif
- Sizes:
    - Title: Large
    - Subheading: Medium
    - Body: Small

Components:

Buttons:

- Rounded corners
- Primary (Green)
- Secondary (Outline)

Cards:

- Soft shadow
- Rounded edges
- Used for:
    - Transactions
    - Balance
    - Reports

Icons:

- Simple line icons
- Finance-related (wallet, chart, money)

Layout Rules:

- Consistent spacing
- Grid-based alignment
- Bottom navigation for easy access

UX Principles Used:

- Simplicity
- Clarity
- Quick access
- Visual feedback (charts)

9. Prototyping & Testing:

After designing all the screens in Figma, an interactive prototype was created. The purpose of this prototype was to simulate how the real mobile application would work.

The prototype connected all screens such as:

- Login & Signup
- Dashboard
- Add Expense
- Transaction History
- Analysis (Daily/Weekly/Monthly)
- Calendar
- Profile

Buttons were made clickable and navigation between screens was added using Figma's interaction feature. This helped in understanding how users will move from one screen to another.

9.1 Prototype

A high-fidelity prototype was developed using Figma. It included realistic UI components such as buttons, cards, charts, and navigation bars.

The prototype allowed users to:

- Log in
- Add new expenses
- View expense categories
- See graphical analysis
- Check calendar expenses
- Navigate using bottom menu

The prototype looked and behaved like a real mobile application.

9.2 Testing & Feedback

The prototype was tested by 5 students and 2 working professionals.

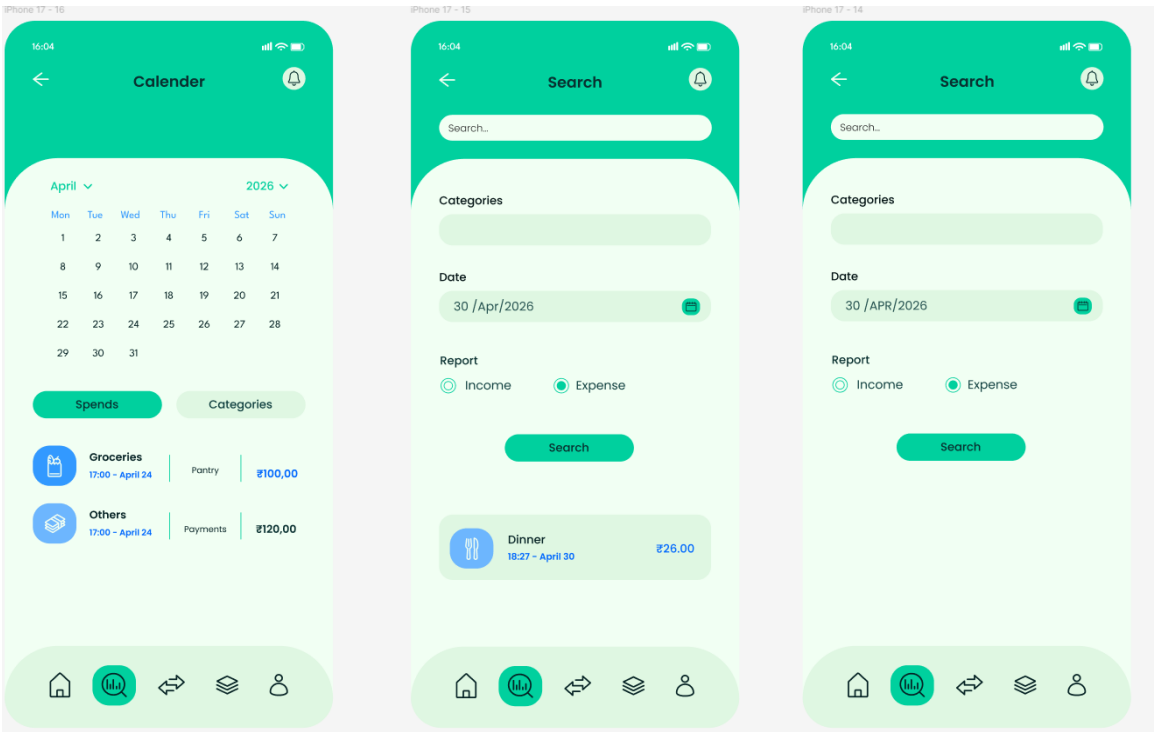
They were asked to:

- Add a new expense
- View monthly analysis
- Check transaction history
- Navigate to profile section

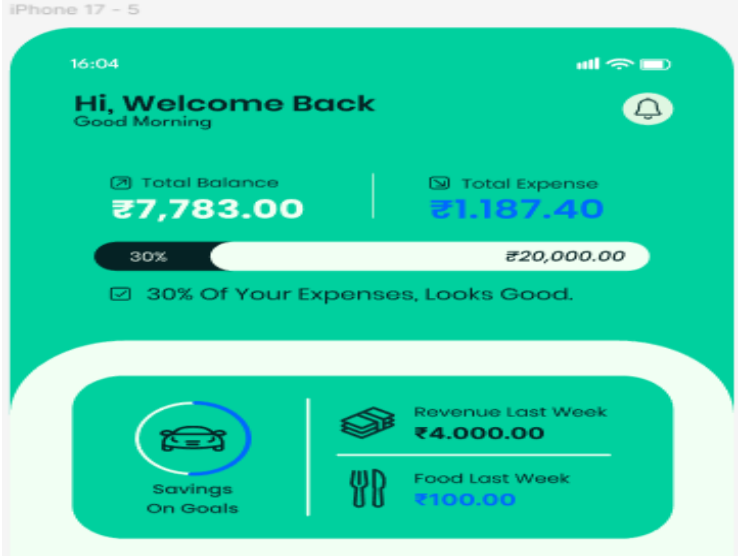
Feedback Received:

Positive Points:

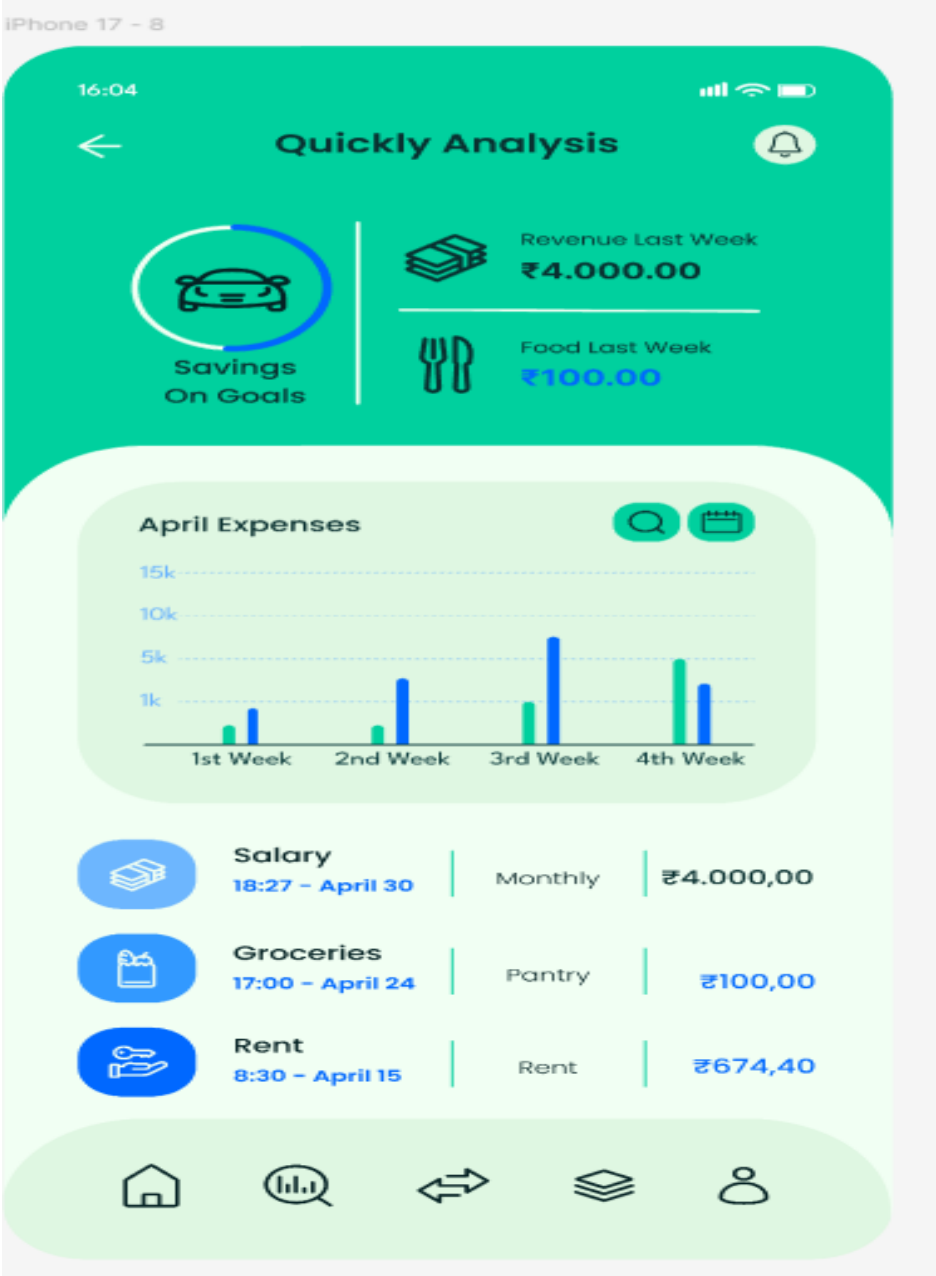
- Clean and simple design



- Clear dashboard balance



- Graphs are helpful



Suggestions for Improvement:

- Increase button size
- Improve graph visibility
- Add confirmation message after saving expense

Based on this feedback, small improvements were made in spacing, button size, and alignment.

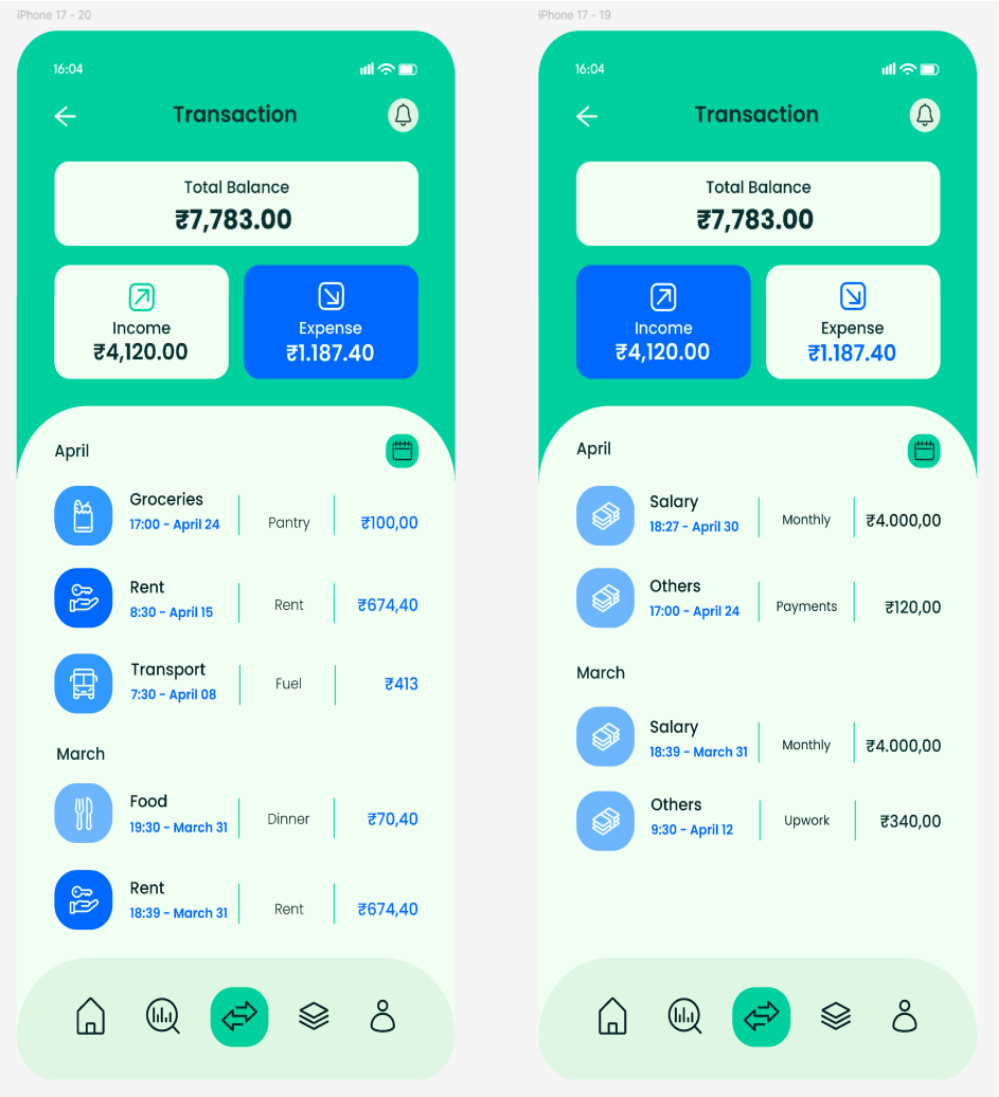
10. Results & Discussion

This section explains what improvements were made and how the design became better.

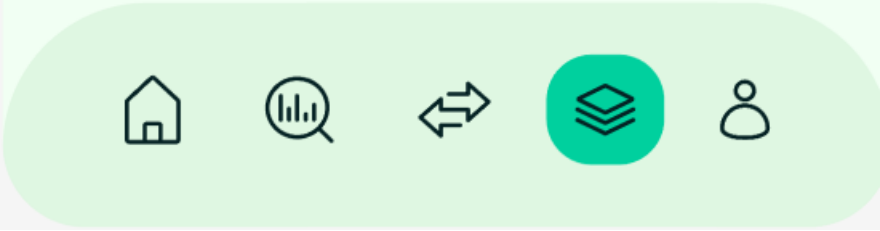
10.1 Improvements Achieved

After testing, the following improvements were made:

- Increased button size for better touch use

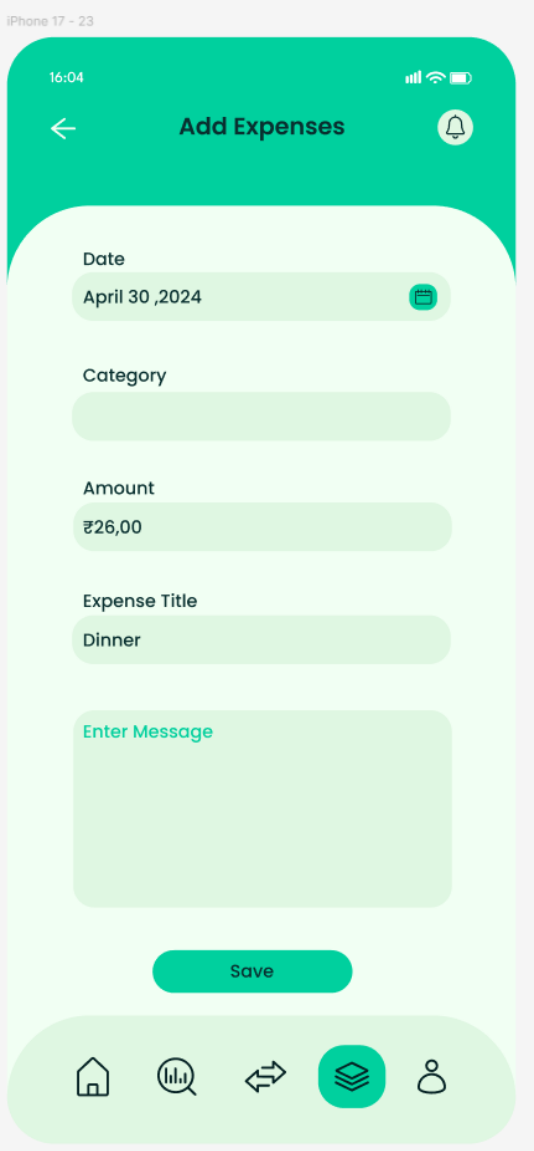


- Improved spacing between dashboard elements



- Enhanced graph readability

- Simplified Add Expense form



These changes made the app more user-friendly.

10.2 User Experience Enhancement

The final design improved user experience in many ways:

- Quick expense entry
- Clear graphical analysis
- Simple bottom navigation
- Clean and modern interface

The application successfully provides a simple and effective way to track expenses.

11. Conclusion

11.1 Summary

The FinWise project was undertaken to address the challenges faced by students and young professionals in managing their daily expenses. Through surveys and stakeholder interviews, the team identified that most users struggle with complex navigation, cluttered dashboards, and slow expense entry in existing applications. To overcome these issues, a user-centered design approach was adopted, focusing on simplicity, clarity, and accessibility. The prototype developed in Figma integrates essential features such as categorized expense cards, dashboard balance overview, transaction history, and graphical analytics. The use of a green-themed interface symbolizes financial growth and stability, while the card-based layout and bottom navigation ensure ease of use. Overall, the project successfully demonstrates how thoughtful UI/UX design can enhance financial awareness and encourage consistent expense tracking.

11.2 Limitations

Despite its strengths, the project has certain limitations. The current scope is restricted to frontend design and prototyping, without backend database integration or real-time synchronization. Features such as payment gateway support, cloud storage, and deployment on Android/iOS platforms were not implemented due to time and resource constraints. Usability testing was conducted on a limited sample size, which may not fully represent diverse user groups or broader demographics. Additionally, advanced financial tools like budget goal tracking and predictive analytics were excluded from the present version, leaving the application as a conceptual prototype rather than a fully functional product.

11.3 Future Scope

The FinWise prototype lays a strong foundation for future development. Upcoming enhancements could include AI-driven expense prediction, personalized budget goals, and expense reminders to improve financial discipline. Integration with cloud services would enable multi-device synchronization and secure data storage, while export functionality could allow users to generate reports in formats like PDF or Excel. Expanding usability testing to a larger and more diverse audience will provide deeper insights into user behavior and help refine the interface further. With backend implementation and advanced analytics, FinWise has the potential to evolve into a comprehensive financial management tool that supports students, professionals, and households in achieving better financial literacy and control.

12:References

- Github Link : https://github.com/Sarthak094/Financial-management-application-/tree/main
- Youtube Link : https://youtu.be/Bs_8xzOpn7A?si=TcEyImTJcAiTY9Nc

- Other references

[1] D. A. Norman, The Design of Everyday Things, Revised and Expanded Edition. New York: Basic Books, 2013.

[2] J. Nielsen, "10 Usability Heuristics for User Interface Design," Nielsen Norman Group, 1994. [Online]. Available: https://www.nngroup.com/articles/ten-usability-heuristics/

[3] B. Shneiderman, C. Plaisant, M. Cohen, S. Jacobs, N. Elmqvist, and N. Diakopoulos, Designing the User Interface: Strategies for Effective Human-Computer Interaction, 6th ed. Boston, MA: Pearson, 2016.

[4] S. Krug, Don't Make Me Think: A Common Sense Approach to Web Usability, 3rd ed. Berkeley, CA: New Riders, 2014.

[5] Google, "Material Design Guidelines," Google Design. [Online]. Available: https://material.io/design

[6] Apple Inc., "Human Interface Guidelines," Apple Developer Documentation. [Online]. Available: https://developer.apple.com/design/human-interface-guidelines/

[7] S. Few, Information Dashboard Design: Displaying Data for At-a-Glance Monitoring, 2nd ed. Burlingame, CA: Analytics Press, 2013.

[8] E. R. Tufte, The Visual Display of Quantitative Information, 2nd ed. Cheshire, CT: Graphics Press, 2001.

[9] Figma, "Figma Help Center and Documentation," Figma Inc. [Online]. Available: https://help.figma.com

[10] J. Gothelf and J. Seiden, Lean UX: Designing Great Products with Agile Teams, 2nd ed. Sebastopol, CA: O'Reilly Media, 2016.

[11] Interaction Design Foundation, “User-Centered Design Principles,” 2023. [Online]. Available: https://www.interaction-design.org

[12] OECD, “Financial Literacy and Financial Education,” OECD Publishing, Paris, 2020.

[13] A. Cooper, R. Reimann, D. Cronin, and C. Noessel, About Face: The Essentials of Interaction Design, 4th ed. Indianapolis, IN: Wiley, 2014.

[14] H. Sharp, Y. Rogers, and J. Preece, Interaction Design: Beyond Human–Computer Interaction, 5th ed. Hoboken, NJ: Wiley, 2019.

[15] D. Benyon, Designing User Experience: A Guide to HCI, UX and Interaction Design, 4th ed. Harlow, UK: Pearson, 2019.

[16] ISO 9241-210, “Ergonomics of Human-System Interaction – Human-Centred Design for Interactive Systems,” International Organization for Standardization, 2019.

[17] B. Moggridge, Designing Interactions. Cambridge, MA: MIT Press, 2007.

[18] L. Rosenfeld, P. Morville, and J. Arango, Information Architecture for the Web and Beyond, 4th ed. Sebastopol, CA: O’Reilly Media, 2015.

13. Appendices

13.1 Survey Questionnaire

The survey was designed to collect both quantitative and qualitative data from students and young professionals aged 18–30. It included multiple-choice questions, Likert scale ratings, and open-ended responses. The questionnaire focused on:

- Frequency of expense tracking
- Preferred analysis methods (graphs vs. numbers)
- Challenges faced in existing apps
- Desired features in a new application

This helped identify user pain points such as lack of simplicity, difficulty in navigation, and preference for quick expense entry.

13.2 Interview Questions

Stakeholder interviews were conducted with engineering students, a working professional, and a faculty member. The structured questions covered:

1. Current methods of expense tracking
2. Problems faced in managing expenses
3. Expected features in an expense tracker
4. Preference for numerical vs. graphical representation
5. Reasons for discontinuing app usage
6. Usefulness of category-based tracking
7. Preference for light or dark mode

The responses highlighted the need for quick entry, simple navigation, category-based classification, and a clean light-themed design.